178065: recrystallized sandstone, Minga Bore

Location and sampling

YOUANMI (SH 50-4), EVERETT CREEK (2841)
MGA Zone 50, 775220E 6849290N

Sampled on 31 October 2002

The sample was taken from a 0.5 m-diameter block about halfway up the southwestern side of a northwest-trending rocky ridge, 50 m southeast of the access track and 0.5 km west of Minga Bore. The sampling site is 50 m west of the site of GSWA sample 178066 (dated by Nelson, this compilation).

Tectonic unit/relations

The sample is a white, coarse- and even-grained, recrystallized quartz sandstone from a bedded quartzite unit in the Maynard Hills greenstone belt of the Southern Cross Granite–Greenstone Terrane, Yilgarn Craton. There is no established stratigraphy for this belt (Riganti, 2003). Bedding in the quartzite unit strikes at 340° and dips steeply to the southwest. The sample was taken 50 m stratigraphically above the unit from which sample 178066 was collected, and contains quartz veins up to 1 mm thick. It was collected to constrain the maximum depositional age of the sandstone precursor to the quartzite.

Petrographic description

The principal mineral is quartz (100 vol.%), with trace amounts of accessory muscovite, rutile, and zircon. The sample consists mostly of interlocking anhedral quartz grains, some of which are optically continuous over areas 10 mm or more in diameter. The quartz contains fluid inclusions. Mica forms flakes 30 to 100 µm in length, and there is a single grain of rutile 100 µm long. The mica defines a weak schistosity, with the quartz c-axes mostly within 45° of the mean orientation of the schistosity. Some mica has been leached, mostly in and adjacent to fractures. The metamorphic grade and protolith are uncertain.

Zircon morphology

The zircons isolated from this sample are typically colourless, pale yellow, dark brown or black, elliptical or subhedral and with rounded terminations, equant or slightly to substantially elongate, and between 50 × 80 µm and 160 × 80 µm in size. Most grains are fractured and internally structureless, but some have faint internal zonation. Many have a speckled appearance in transmitted light, and many are metamict. Many grain surfaces are pitted, consistent with detrital transport. In reflected light, many surfaces have irregular, dark patches and zones. Cathodoluminescence images of representative zircons are given in Figure 1.

Analytical details

This sample was analysed on 5 May 2003. The counter deadtime during the analysis session was 24 ns. Fourteen analyses of the CZ3 standard obtained during the analysis session indicated a Pb*/U calibration uncertainty of 1.39% (1σ). Common-Pb corrections were applied assuming Broken Hill common-Pb isotopic compositions for all analyses, with the exception of analyses 8.1 and 12.1, for which isotopic compositions determined using the method of Cumming and Richards (1975) were assumed.

Results

Thirty-nine analyses were obtained from 39 zircons. Results are given in Table 1, and shown on concordia and Gaussian-summation probability density plots in Figures 2 and 3 respectively.

Interpretation

The analyses are concordant to highly discordant, with the discordance pattern consistent with both ancient and recent episodes of radiogenic-Pb loss. On the basis of their $^{207}Pb/^{206}Pb$ ratios, many analyses can be assigned to one of six groups. Eight concordant to highly discordant analyses of eight zircons (2.1, 4.1, 6.1, 9.1, 20.1, 22.1, 29.1, 36.1), assigned to Group 1, have $^{207}Pb/^{206}Pb$ ratios defining a single population and indicating a weighted mean $^{207}Pb/^{206}Pb$ date of 3276 ± 7 Ma (chi-squared = 1.57). Concordant analyses 13.1, 16.1, and 28.1, and discordant analysis 30.1, assigned to Group 2, have $^{207}Pb/^{206}Pb$ ratios defining a single population and indicating a weighted mean $^{207}Pb/^{206}Pb$ date of 3324 ± 15 Ma (chi-squared = 1.27). Thirteen concordant to highly discordant analyses of 13 zircons (1.1, 3.1, 5.1, 7.1, 10.1, 14.1, 17.1, 19.1, 26.1, 31.1, 33.1, 37.1, 39.1), assigned to Group 3, have $^{207}Pb/^{206}Pb$ ratios defining a single population and indicating a weighted mean $^{207}Pb/^{206}Pb$ date of 3379 ± 6 Ma (chi-squared = 1.59). Concordant analyses 27.1 and 35.1, and discordant analysis 25.1, assigned to Group 4, have $^{207}Pb/^{206}Pb$ ratios defining a single population and indicating a weighted mean $^{207}Pb/^{206}Pb$ date of 3419 ± 14 Ma (chi-squared = 0.70). Concordant analysis 34.1 and discordant analyses 15.1 and 32.1, assigned to Group 5, have $^{207}Pb/^{206}Pb$ ratios defining a single population and indicating a weighted mean $^{207}Pb/^{206}Pb$ date of 3491 ± 19 Ma (chi-squared = 0.75). Concordant analysis 23.1 and discordant analyses 11.1 and 12.1, assigned to Group 6, have $^{207}Pb/^{206}Pb$ ratios defining a single population and indicating a weighted mean $^{207}Pb/^{206}Pb$ date of 3596 ± 19 Ma (chi-squared = 1.58). The remaining analyses cannot be confidently grouped.

The analyses are interpreted to be of detrital zircons. The date of 3276 ± 7 Ma indicated by the weighted mean

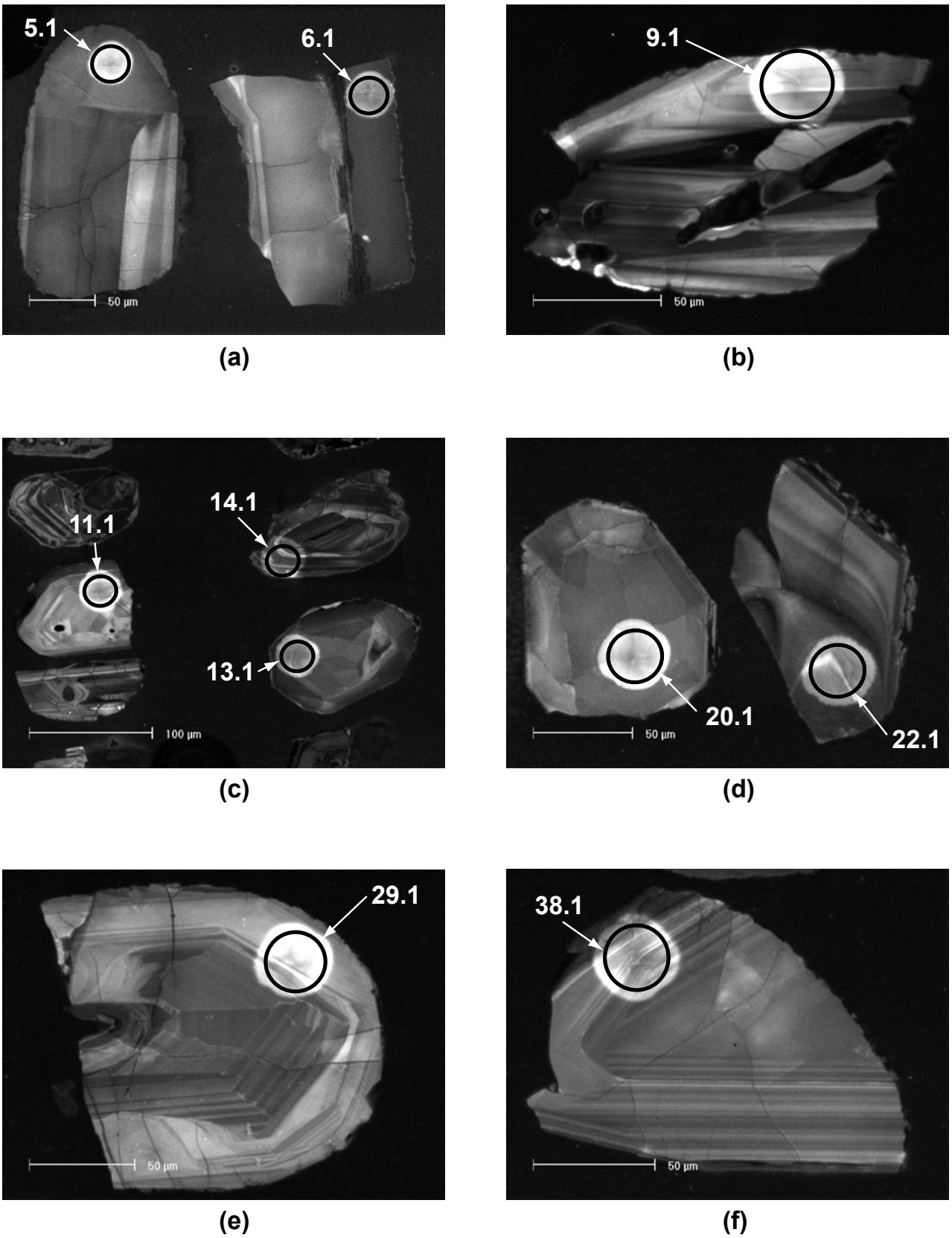

Figure 1. **Cathodoluminescence images of representative zircons from sample 178065: recrystallized sandstone, Minga Bore**

Table 1. Ion microprobe analytical results for zircons from sample 178065: recrystallized sandstone, Minga Bore

Grain .spot	*U (ppm)*	*Th (ppm)*	*Pb (ppm)*	*f206%*	$^{207}Pb/^{206}Pb$	±1σ	$^{208}Pb/^{206}Pb$	±1σ	$^{206}Pb/^{238}U$	±1σ	$^{207}Pb/^{235}U$	±1σ	*% concordance*	$^{207}Pb/^{206}Pb$ *Age*	±1σ
1.1	156	245	112	0.410	0.28132	0.00147	0.03927	0.00183	0.6229	0.0093	24.162	0.396	93	3 370	8
2.1	575	572	479	0.015	0.26465	0.00061	0.26181	0.00080	0.6338	0.0090	23.129	0.338	97	3 275	4
3.1	193	69	157	0.066	0.28465	0.00111	0.09327	0.00102	0.6863	0.0101	26.936	0.422	99	3 389	6
4.1	234	482	235	0.108	0.26304	0.00105	0.50255	0.00209	0.6563	0.0096	23.804	0.371	100	3 265	6
5.1	100	63	87	0.061	0.28626	0.00152	0.16668	0.00168	0.6899	0.0106	27.231	0.458	100	3 397	8
6.1	197	418	206	0.093	0.26603	0.00108	0.56064	0.00226	0.6622	0.0097	24.289	0.380	100	3 283	6
7.1	185	106	150	0.150	0.28120	0.00124	0.15616	0.00154	0.6518	0.0096	25.271	0.401	96	3 370	7
8.1	818	510	534	0.142	0.23906	0.00053	0.15223	0.00065	0.5422	0.0076	17.871	0.260	90	3 114	4
9.1	47	48	40	0.199	0.26155	0.00235	0.26690	0.00368	0.6523	0.0111	23.523	0.475	99	3 256	14
10.1	180	80	148	-0.015	0.28220	0.00130	0.11553	0.00178	0.6851	0.0101	26.658	0.426	100	3 375	7
11.1	72	67	71	0.106	0.32429	0.00201	0.24250	0.00269	0.7264	0.0115	32.478	0.576	98	3 590	10
12.1	474	293	346	0.305	0.32467	0.00083	0.17105	0.00103	0.5615	0.0080	25.137	0.370	80	3 592	4
13.1	118	126	107	-0.024	0.27497	0.00162	0.28620	0.00272	0.6773	0.0103	25.678	0.435	100	3 335	9
14.1	266	327	226	0.349	0.28224	0.00106	0.31443	0.00174	0.6111	0.0088	23.783	0.365	91	3 375	6
15.1	158	98	129	0.177	0.30447	0.00133	0.16588	0.00154	0.6403	0.0095	26.879	0.429	91	3 493	7
16.1	129	161	119	-0.008	0.27217	0.00131	0.32824	0.00200	0.6711	0.0101	25.183	0.410	100	3 319	8
17.1	142	94	116	0.061	0.28227	0.00127	0.16682	0.00141	0.6532	0.0097	25.421	0.408	96	3 376	7
18.1	346	344	259	0.132	0.26204	0.00083	0.26410	0.00121	0.5666	0.0081	20.469	0.307	89	3 259	5
19.1	15	7	14	0.553	0.29803	0.00645	0.12036	0.01125	0.7279	0.0165	29.912	0.996	102	3 460	34
20.1	100	90	83	0.003	0.26521	0.00141	0.23983	0.00172	0.6404	0.0098	23.418	0.392	97	3 278	8
21.1	240	230	182	0.354	0.27797	0.00113	0.16492	0.00157	0.6037	0.0088	23.136	0.359	91	3 352	6
22.1	214	86	160	0.090	0.26466	0.00104	0.09628	0.00108	0.6364	0.0093	23.223	0.361	97	3 275	6
23.1	204	47	182	0.037	0.32743	0.00114	0.06076	0.00098	0.7449	0.0109	33.629	0.520	100	3 605	5
24.1	142	81	140	0.073	0.34633	0.00137	0.14555	0.00128	0.7631	0.0114	36.439	0.580	99	3 691	6
25.1	135	95	114	0.219	0.28899	0.00142	0.19236	0.00182	0.6609	0.0099	26.333	0.429	96	3 412	8
26.1	49	75	49	-0.111	0.28260	0.00420	0.40670	0.00892	0.6899	0.0117	26.882	0.643	100	3 377	23
27.1	83	67	75	0.170	0.28932	0.00172	0.20875	0.00220	0.6986	0.0108	27.869	0.480	100	3 414	9
28.1	149	134	130	0.054	0.27438	0.00121	0.23528	0.00154	0.6721	0.0100	25.426	0.407	99	3 331	7
29.1	42	64	41	0.363	0.26683	0.00281	0.38872	0.00538	0.6786	0.0115	24.966	0.524	102	3 288	17
30.1	158	142	130	0.118	0.27131	0.00133	0.23176	0.00200	0.6325	0.0094	23.663	0.383	95	3 314	8
31.1	84	36	70	0.093	0.28239	0.00168	0.11433	0.00172	0.6955	0.0108	27.082	0.470	101	3 376	9
32.1	95	28	77	0.289	0.30184	0.00181	0.05305	0.00219	0.6910	0.0107	28.758	0.496	97	3 480	9
33.1	253	43	189	0.166	0.28362	0.00100	0.02541	0.00088	0.6589	0.0095	25.768	0.394	96	3 383	6
34.1	117	21	97	0.118	0.30489	0.00154	0.04672	0.00153	0.7130	0.0108	29.975	0.497	99	3 495	8
35.1	411	425	384	0.062	0.29087	0.00072	0.25539	0.00091	0.6997	0.0100	28.062	0.414	100	3 422	4
36.1	211	226	184	0.112	0.26733	0.00107	0.28671	0.00162	0.6518	0.0095	24.026	0.374	98	3 290	6
37.1	89	42	74	0.055	0.28060	0.00162	0.12627	0.00169	0.6862	0.0106	26.548	0.457	100	3 366	9
38.1	104	98	87	0.100	0.27747	0.00154	0.18338	0.00195	0.6568	0.0100	25.127	0.423	97	3 349	9
39.1	126	109	95	0.124	0.28420	0.00143	0.06857	0.00139	0.6440	0.0096	25.236	0.413	95	3 386	8

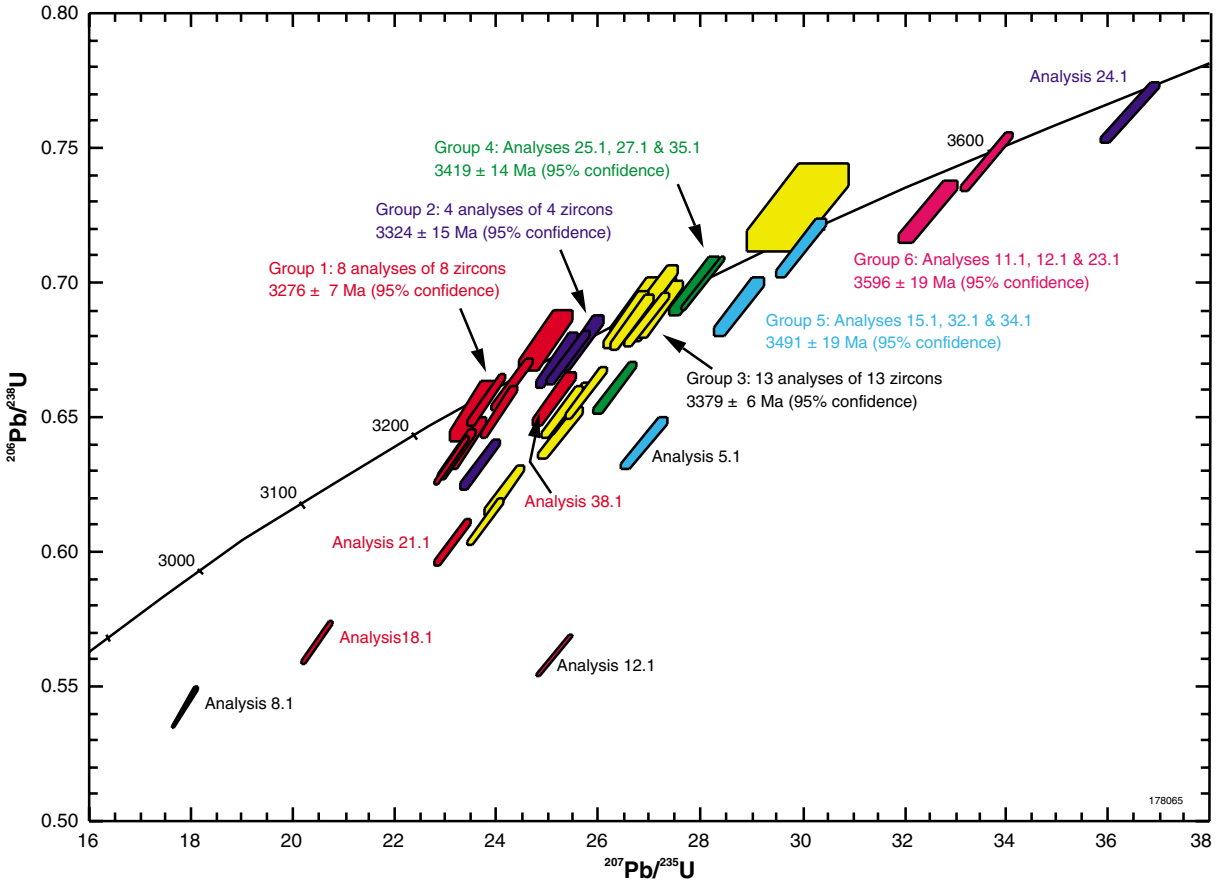

Figure 2. Concordia plot for sample 178065: recrystallized sandstone, Minga Bore

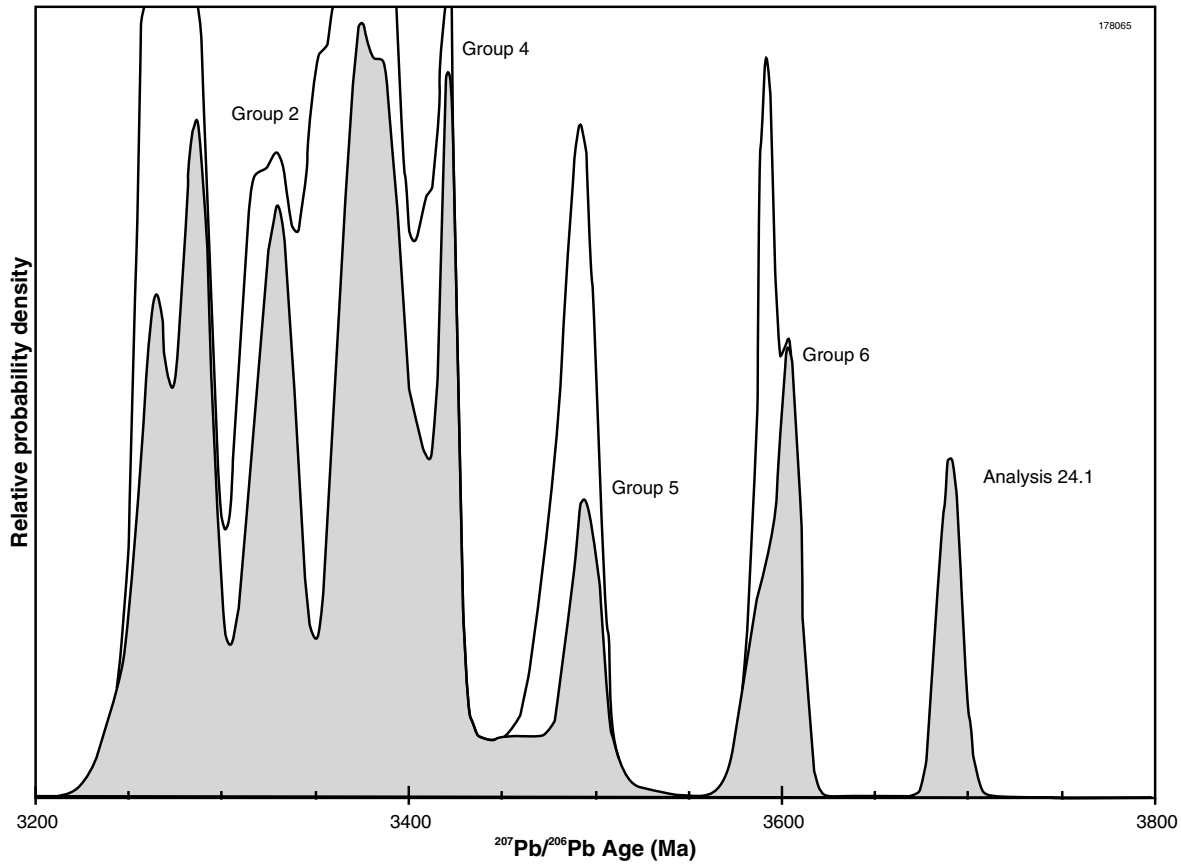

Figure 3. Gaussian-summation probability density plot for sample 178065: recrystallized sandstone, Minga Bore

$^{207}Pb/^{206}Pb$ ratio of eight concordant to highly discordant analyses of Group 1 is therefore interpreted as providing a maximum age for deposition of the precursor sandstone. Source rocks within the western part of Australia having ages matching those of the zircons within this sample include the Narryer Terrane in the northwestern part of the Yilgarn Craton (Nelson, 1995, 1996, 1997) and the eastern part of the Pilbara Craton (Nelson, 1996, 1997, 1998, 1999, 2000, 2001, 2002, this compilation).

Recommended reference for this publication:

NELSON, D. R., 2005, 178065: recrystallized sandstone, Minga Bore; Geochronology dataset 592; in Compilation of geochronology data, June 2006 update: Western Australia Geological Survey.

Data obtained: 05/05/2003; Data released: 30/06/2005